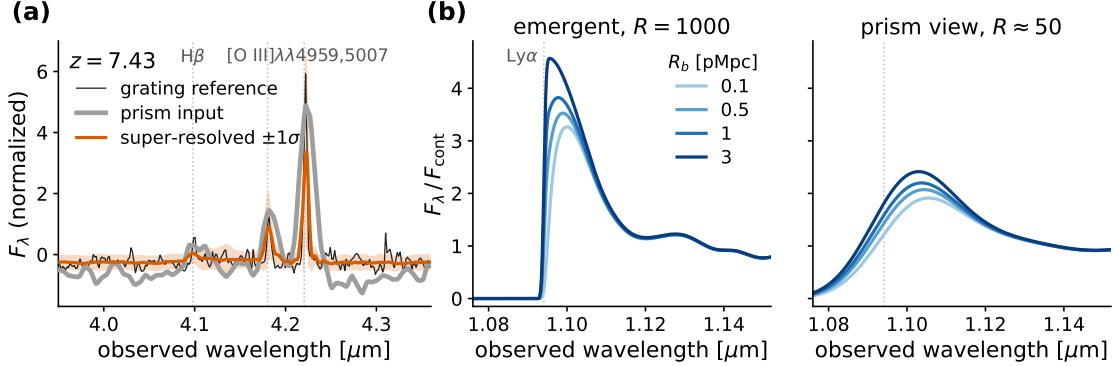


Deep Learning for Galaxy Spectral Super-Resolution

Aryana Haghjoo

Modern infrared telescopes like JWST and Nancy Grace Roman enable cosmic observations [3], but efficient spectroscopic modes operate at low resolution: JWST’s NIRSpec prism ($R \sim 100$) [6] and Roman’s grism. Spectral features encoding critical physics blur below instrumental limits. I develop physics-informed deep learning frameworks that super-resolve low-resolution galaxy spectra by $\sim 10\times$, recovering hidden astrophysical information [4].

Using paired JWST/NIRSpec observations from JADES [1] (same galaxies in high-resolution and prism modes), I train generative models with physics-informed loss functions [7, 12, 5]. Figure 1(a) demonstrates super-resolution on a $z = 7.43$ galaxy, recovering features invisible in prism data. Figure 1(b) shows the astrophysical impact: Lyman-alpha profiles encode ionized bubble sizes (R_b) around galaxies. High-resolution profiles are distinguishable; prism-resolution profiles appear identical. Our method recovers this hidden information, enabling physical measurements from scientifically limited data.



Summer 2027 at IPAC: The fellowship focuses on four integrated components: (1) validating recovery of astrophysical diagnostics using held-out JADES data, comparing deep learning against classical deconvolution methods [8, 9, 11]; (2) adapting the framework to Roman grism with IPAC’s mission team; (3) developing multi-modal models; (4) extending to Euclid spectroscopy. Timing is critical: Roman’s first data release is expected Summer 2027. On-site access to real mission data and Dr. Shoubaneh Hemmati—active member of Euclid and Roman missions—enables immediate validation impossible remotely. In-person collaboration ensures our methodology aligns with mission requirements and delivers forecasts to inform survey strategy.

This work advances galaxy astrophysics while demonstrating how machine learning extracts maximum value from observational data [10, 2]. Open-source code and models will enable researchers to apply spectral super-resolution across datasets, extending the science reach of JWST, Roman, Euclid, and future missions.

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